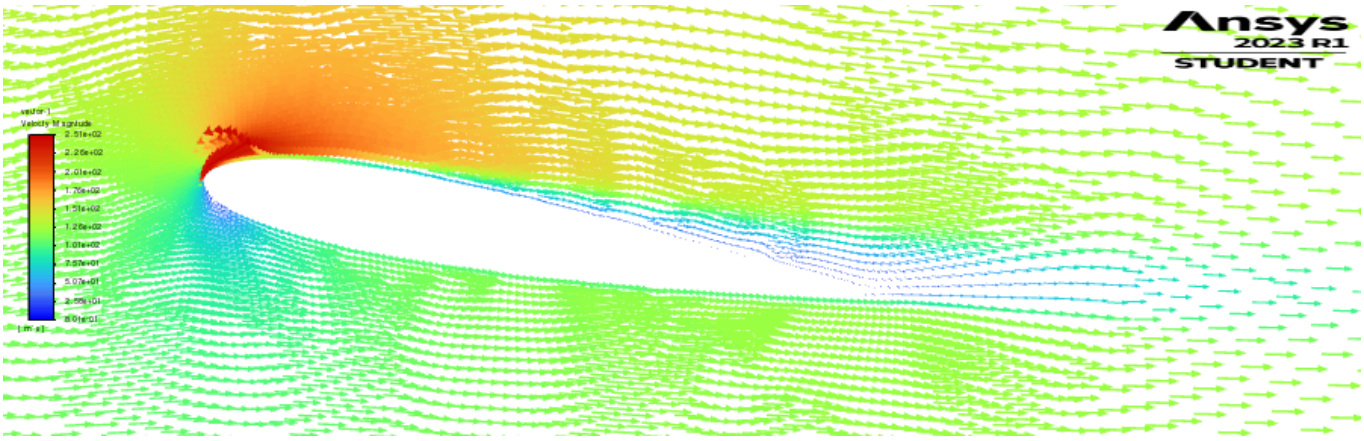
MMME3086: COMPUTER MODELLING TECHNIQUES

CFD ASSIGNMENT

10 Credits from a 20 credit module

Set: 29th October 2026 | Due: 12th November 2026

Staff Contact: Dr Donald Giddings



MODULE CODE	MODULE TITLE	
MMME3086	*Computer Modelling Techniques*	
ASSESSMENT TITLE		
CFD assignment		
ASSESSMENT TYPE		**WEIGHT (INDICATIVE EFFORT)**
Individual Coursework		*10* credits 20 hours
DEADLINE DATE	**DEADLINE TIME**	**SUBMISSION METHOD**
Thursday 12th November 2026	15:00	*Moodle submission box*
STAFF MEMBER RESPONSIBLE		
Dr Donald Giddings Donald.giddings@nottingham.ac.uk		
FEEDBACK DETAILS		
A mark, and comments on Moodle will be provided by Thursday 3rd December 2026		

LEARNING OUTCOMES ASSESSED

LO1	***Understand the theoretical background of numerical methods used in engineering analysis, Finite Element and Computational Fluid Dynamics techniques. - AHEP4: 1, 2***
LO2	***Identify and apply appropriate computer modelling techniques to solve specific engineering problems. - AHEP4: 3***
LO3	***Demonstrate an understanding of how computer programmes can be used to solve practical solid and fluid mechanics problems in engineering. - AHEP4: 3***
LO4	***Evaluate the accuracy and sources of error in solving solid and fluid mechanics problems in engineering using computational models. - AHEP4: 2***
LO5	***Gain hands-on experience of running state of the art FEA and CFD software codes widely used in industry - AHEP4: 13***

KEY SUBMISSION REQUIREMENTS

- *An 8 page report with strict length limits*
- *Graphic and text representation of ideas communicated*

NO EVIDENCE EC ELIGIBLE?[1]	EC OUTCOME
(No)	*Extension or First sit depending situation*

Before submitting an EC, please discuss with your tutor.

SUPPORT PLAN EXTENSION[2]

If you have an existing support plan *and* require a support plan extension for this assessment, you should email EZ-M3-DLO@exmail.nottingham.ac.uk. Please include module code, assessment name, original deadline including time, and requested extension (usually one working week).

MISCONDUCT INFORMATION[3]

[1] EC regulations: https://www.nottingham.ac.uk/qualitymanual/assessment-awards-and-deg-classification/ext-circumstances.aspx

[2] Support plans: https://www.nottingham.ac.uk/studentservices/servicedetails/disability-support-services/disability-support-services.aspx

[3] Details of the University's policy procedures on Academic Misconduct and disciplinary procedures:

| *Context*

The learning outcome(s) of the coursework assessment is to develop a 2D CFD model that represents the flow around an aerofoil in a duct. This will help you to develop aspects of all the learning outcomes of the module.

The CFD assessment provides an opportunity to learn how to use a commercially available CFD software package. You will set up the model, solve the fluid flow and analyse the data produced. The procedure will involve establishing that a proper CFD investigation has been done including mesh dependence check, numerical convergence and analysis of physical representation according to validation data available in the literature.

| Building on Previous work

This assessment builds upon knowledge, skills, and feedback gained throughout your degree programme. Before starting the assessment, you are encouraged to review the following:

- **Fluid mechanics modules in year 1 and year 2**

| Supporting your future development

The knowledge, skills, and feedback gained will support your development in future assessments and learning opportunities, including *project work as appropriate and further analysis of any fluid flow situation.*

| *Workload breakdown*

This assessment is worth 10 credits of a 20 credit module – ~20 hours of work to complete.

| *Use of Artificial Intelligence*

| Declaration of AI usage

A log: Students must provide a summary of the AI usage log in an appendix, maximum [2] pages, addressing all seven points in the AI usage log template.

| Guidance on AI usage

All assessments must include an AI guidance table.

Detailed guidance on AI usage is provided in the table in Appendix 1

| *Submission information*

Naming File: Last-name_First-name_Student-ID-number_CFDAssignment.PDF

File Type Required: PDF

Title of Submission box on Moodle: CFD Assignment Submission box

Students must submit their work using the correct naming convention, and in the file type listed above. Failure to do this may result in penalties if staff are unable to make an appropriate judgement on your work due to incorrect file types.

Your submission deadline is 3 pm on 12th November 2026.

You should use the template provided for your submission, completing the Word document available on Moodle. Please do not change the format of the document. You can print the relevant pages directly to pdf to be uploaded. There are word counts for specific parts of the assignment, please observe the word count so that you complete a precise and concise report.

It is your responsibility to check the file you have uploaded to ensure that it has uploaded correctly.

Late submission shall incur the normal university penalties: Any assessment uploaded after 15:00:00 on the submission date will be counted as late and will receive a 5% penalty. An additional 5% penalty will be applied for each additional working day that it is late. **Please ensure you upload your work well prior to the submission deadline to ensure upload issues do not create an unexpected late penalty.** Similarly, it is important that you read through all the documentation for this assessment and ask any questions in plenty of time ahead of the deadline. Raising an issue the day or an hour before the deadline is not a suitable time frame. We recommend storing your final document on OneDrive in a secure location to preserve the date of completion in case of unpredictable complications upon submission.

If you have submitted an EC but have not had a response from the panel at the time of the deadline, **you must submit the current work you have completed, even if this is not finished, with the words EC PENDING in the title**. You should then:

- notify the staff member responsible for the assessment that you have submitted an EC and your proposed deadline (do not exceed the maximum extension for an EC as stated on the moodle page)
- continue to work to, and submit by, the extension date requested in your EC claim.

If you are then granted an extension, your later submission will be marked.

Brief

Date issued:	Thursday 29/10/2026 , 15:00
Deadline for submission:	Thursday 12/11/2026, 15:00
Submission format:	- Online only on Moodle (not a paper copy) using the template provided on Moodle, the format is figures with captions and tables and indicated text commentary boxes. Each item of text (caption or text box) should be **no greater than the stated word count**. Obvious overlength will have a penalty for inefficient communication. Save as pdf.
Marks allocated:	50% of the MMME3086 module mark
How to access Ansys Workbench:	- Ansys Workbench is available on the Virtual Engineering Desktop, this is the 2026 R1 version. - Note: You may download a free "Student Edition" of Ansys for home use: https://www.ansys.com/en-gb/academic/free-student-products - Ansys Workbench is also available in departmental computer rooms
Timetabled sessions:	- This coursework assignment does not require supervision. There are no timetabled computer sessions, but there is the 2nd hour of the Thursday practical sessions in which support tutors will assist with specific questions.
Estimated time to be spent on this assignment:	- Approximately 20 hours

Feedback and Marks

Individual feedback will be provided on 03/12/2026. This will be available on Moodle in the submissions Grade box in the form of a mark and an assessment rubric with generic and specific commentary. An announcement will be made when feedback and marks are available.

Task (A) – generating a geometry and mesh [16% of coursework marks]

The purpose of this task is to gain some hands-on experience of setting up a geometry using Ansys Discovery and various computational meshes in Ansys Mesher.

Using the step-by-step consolidation method (available on Echo360)

A1. Generate your own 150 mm by 1 m duct with a NACA0012 aerofoil at an **8°** angle of attack. Illustrate the geometry with a short descriptive caption in Figure 1. **[4]**

A2. Generate and illustrate in Figure 2 a mesh with 10 mm triangular cells on the duct, but with 2 mm cell Sizing associated with the walls of the **aerofoil only** and a growth rate of 1.05 of the cells away from the aerofoil. In the caption (<100 words) add a comment on your consideration of the model about the CFD theory of the mesh. Display the local region around the aerofoil in an inset image. **[8]**

A3. Add appropriate names to all of the boundaries of the model and label them in your figure from part A2. **[4]**

Task (B) – generating boundary conditions and solving

[34% of coursework marks]

Set up a case using the aerofoil mesh you have generated. Set the velocity inlet magnitude to the value (n/100)×10+50 m/s, where n is the number formed by the last two digits of your student ID number (should produce Re = O(10^5) based on chord length of 100 mm) and use the default pressure outlet boundary condition. On models, select under viscous model the settings for each of the 5 cases described in sections B1 and B2 below.

B0. Note in your report **[6]**:

a. density of air
b. viscosity of air
c. turbulence intensity and isotropic RMS velocity of turbulent fluctuations at inlet
d. velocity at inlet
e. calculated Reynolds number based on chord length

B1. **Case 1**: Use the ***Spalart-Allmaras*** *one equation turbulence model using* ***standard wall function*** *together with the* ***SIMPLE*** *pressure-velocity coupling method*. Initialise the case and set the calculation and run it until convergence is achieved. Print a graphic, in Figure 3, showing the residual history and note in the caption what happened to the convergence of the case and what you think it might indicate. **[5]**

Note the number of iterations to convergence. From the data, extract to an Excel spreadsheet the following:

1. static pressure and y+ on suction side,
2. static pressure and y+ on pressure side.
3. Insert three lines, across the duct, one at mid chord, one at trailing edge and one 100 mm downstream from the trailing edge. Extract to Excel the velocity magnitude for the three lines.
4. Record the total drag and lift on the aerofoil and the convergence criterion and iterations to convergence.
5. Record this data for all subsequent models also.

Case 2: Set up a new case with the same mesh by the 'Duplicate' method in Workbench. Run it with ***k-ε turbulence model***, ***standard wall function*** and **SIMPLE** pressure-velocity coupling and record the data for this model.
Case 3: Duplicate the setup again, and run the ***RSM model***, *with* ***standard wall function***, *and with* ***SIMPLE*** *pressure-velocity coupling* method. Record the data.

B2. **Results analysis**: For the RSM **Case 3**, present a vector plot around the aerofoil showing the wake region clearly in Figure 4. In the caption state what it is you are showing and what it is that you see in the flow pattern and state the lift and drag for this case. State the lift and drag forces and the iterations to convergence and the convergence criterion. **[6]**
In Figure 5, present a contour plot of y+ values on the cells in touch with the aerofoil for **Case 3**, showing the cell values and showing the legend to indicate range of values of the cells in contact with the aerofoil. Indicate any unusual regions stating why you have highlighted them with reference to the wall function that was applied. **[5]**

B3. Load the mesh Aerofoil8-degAoA-enhancedBL.msh into Fluent directly. Run the **Case 4** with **RSM** and **enhanced wall treatment** with **SIMPLE**, state what you observe. Run the **Case 4** again with **SIMPLE**, **k-ε** turbulence model, and use **enhanced wall treatment** in the turbulence model. Note the convergence rate and the drag and lift forces and record its data as for the models in B1. Duplicate and set up a new **Case 5** to change the turbulence method back to **RSM**, alter the **under-relaxation** on Reynolds stresses to 0.25, all other turbulent and momentum under-relaxation parameters to 0.5, and then continue to run the case until convergence or until it doesn't change. Collect the data. **Report** the drag and lift forces and describe how the residuals behave for each of Case 4 and Case 5 and comment what changed between the two RSM cases here. **[6]**.
Present a graphic in Figure 6 containing both a graphic showing the velocity vectors and another graphic of the y+ for the **RSM** case and describe in the caption what you can see and interpret the outcome with respect to CFD theory. **[6]**

Task (C) – using post processing skills and turbulence models [50% of coursework marks]

The purpose of this part is to demonstrate ability to analyse the results from a CFD model.

For part C1, C2 and C3, Consider the models used and the meshes they were used on and make brief statements especially regarding the reliability of the methods used when you state your observations. Each statement should be less than 120 words long.

C1. Compare the velocity data on the three lines for all models in 3 graphs and make briefly state your observations in less than 100 words for each figure.

[10]

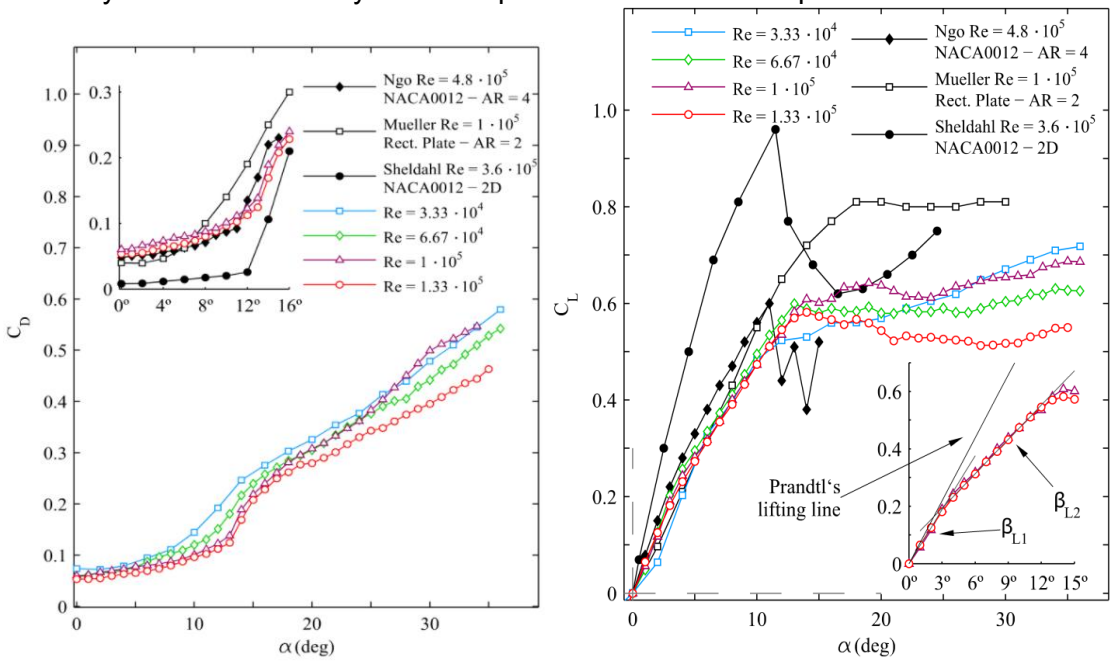
C2. Compare the pressure data on the pressure side of the aerofoil and the suction side for all five cases on two graphs and briefly state your observations in no more than 100 words about each figure.

[10]

C3. Compare the y+ values on the pressure and suction sides of the aerofoil for all the models and briefly state your observations in no more than 100 words combined.

[10]

C4. Compare the 8° angle of attack C_D, and C_L with the Sheldahl result in the figure below. Comment on your Reynolds number compared with that of Sheldahl and your model with the physical setup of the model of Sheldahl. Comment, in less than 550 words, on the accuracy of your own simulations versus the experimental values and discuss what the effects of the CFD models should be in producing better reliability. Comment on why it is so important to match the experimental value.



Use the planform area, assuming 1 m depth, to determine the force. Notice the definition of the tests done by Sheldahl (in the original paper, the aerofoil chord is 150 mm and the span is 900 mm, and the wind tunnel cross section is 2.1 m high) and comment on how well they compare to the boundary conditions of the 2D model in CFD. **[20]**

You can review the paper at:
http://article.sapub.org/10.5923.j.aerospace.20160401.01.html[4]

[4] S. Martínez-Aranda, A. L. García-González, L. Parras, J. F. Velázquez-Navarro, C. del Pino, 2016, Comparison of the Aerodynamic Characteristics of the NACA0012 Airfoil at Low-to-Moderate Reynolds Numbers for any Aspect Ratio, International Journal of Aerospace Sciences, 4(1): 1-8.

Assessment Criteria

See Excel file for improved visibility if required.

Part	Elements for Consideration	Weight	Poor	Less than Acceptable	Acceptable	Good	Excellent	Outstanding
A	Demonstrate ability with geometry construction. Present figures showing the required developments on each part with good consideration of the presentation and with descriptive captions of appropriate length. Format of template is used appropriately.	20	Poor attempt at the skills required is presented or the report is very poorly presented with problems on figures or captions or understanding or a combination of each. Not enough work to assess well.	The skills may not be well developed or well presented or well described, or a product of problems inconsistent across the aspects required.	The skills required have been well attempted generally but may have some errors or omissions or displaying limitations and the descriptions in the captions may be improved by simple changes.	The skills required are demonstrated with perhaps some errors in understanding the task or in confusion over presentation and description. Overall well presented and described reasonably well.	All skills on geometry and mesh are generated and demonstrated. Captions describe the developments well. All elements well described and presented in appropriate format.	An exceptional approach to the work demonstrating a highly skilled practice with the CFD software and an excellent understanding of aspects of the graphics obtained together with outstanding description and presentation.
		Marks	0-39	40-49	50-59	60-69	70-79	80-100
		Award						
B	Demonstrate ability with presenting CFD data. Good illustration of figures and descriptive captions. Good consideration of CFD aspects observed from the models.	30	Poor attempt at the skills required is presented or the report is very poorly presented with problems on figures or captions or understanding or a combination of each. Not enough work to assess well.	The skills may not be well developed or well presented or well described, or a product of problems inconsistent across the aspects required.	The skills required have been well attempted generally but may have some errors or omissions or displaying limitations and the descriptions in the captions may be improved by simple changes.	The skills required are demonstrated with perhaps some errors in understanding the task or in confusion over presentation and description. Overall well presented and described reasonably well.	All skills on geometry and mesh are generated and demonstrated. Captions describe the developments well. All elements well described and presented in appropriate format.	An exceptional approach to the work demonstrating a highly skilled practice with the CFD software and an excellent understanding of aspects of the graphics and data obtained together with outstanding description and presentation.
		Marks	0-39	40-49	50-59	60-69	70-79	80-100
		Award						
C	Demonstrate acquisition of CFD skills as required in the tasks and the post-processing required. Present appropriately clearly with considered brief captions and text explaining the figures. Correct understanding of the aspects asked and questions posed.	50	There is little or no coherent approach to the tasks and presentation. This generally reflects insufficient work done for some reason.	Attempted work has aspects that require improvement in various parts. Although some skills are acquired and some attention to presentation is demonstrated, the overall impression is that things can be improved in general.	There is a good attempt at all aspects and some areas for improvement in the demonstration of the required points and few areas for improvement. The graphics are clear but have room for improvement, the data statements are mostly correct and the captions support the figures reasonably well.	The work is attempted in every aspect with good presentation of figures, and captions capturing most of the description and data that is correctly acquired. There is room for some improvement on areas of presentation or understanding or skills demonstrated in data acquisition.	Required data, graphics and captions are well generated with clear demonstration of underlying understanding and skills acquired. Figures and captions are well presented with only small areas for improvement. Knowledge and understanding of the acquired CFD data are nearly entirely demonstrated with some small areas for improvement in presentation or combination of aspects.	All required data, figure generation and descriptive captions demonstrate clearly high level of skill and understanding in using the CFD. Figures and captions are complete and knowledge and understanding of the required tasks is clearly demonstrated with little or no room for improvement. Aspects of style that elevate the mark are apparent together with exceptional overall approach.
		Marks	0-39	40-49	50-59	60-69	70-79	80-100
		Award						
	Total	100						